

UBL figures: a uniform set

A demonstration for the TC, not a decision. Five of the 78 activity diagrams, from the simplest to the most complex, shown three ways.

1. **As published today.** Each figure is the committed SVG, faithful to its artwork, and shown at the column's full width, as the specification shows every figure. So a figure with a few boxes has large boxes and large text, and a figure with many has small ones.
2. **In a house style, fitted to the column.** The same element looks the same in every figure: one label size, one line weight, one arrowhead, one shape for each kind of node. It is still shown at the column's width, so its size on paper still depends on how much the figure holds: its labels print at exactly the size they do today, since each figure keeps its layout and is scaled as a whole.
3. **In the house style, at one scale for all.** Every figure is printed at the same scale, so a label is the same size on paper (7 pt) in every figure. A simple figure is small; a complex one needs a larger page, or a new layout that fits the page.

In draw.io this is natural. A draw.io page is not a fixed size: the drawing is made in its own units, the same for every figure, and the page and the export scale decide how large it prints. What the TC has to choose is which of the two kinds of uniform it wants on paper: the same *look* (2), or the same *size* as well (3).

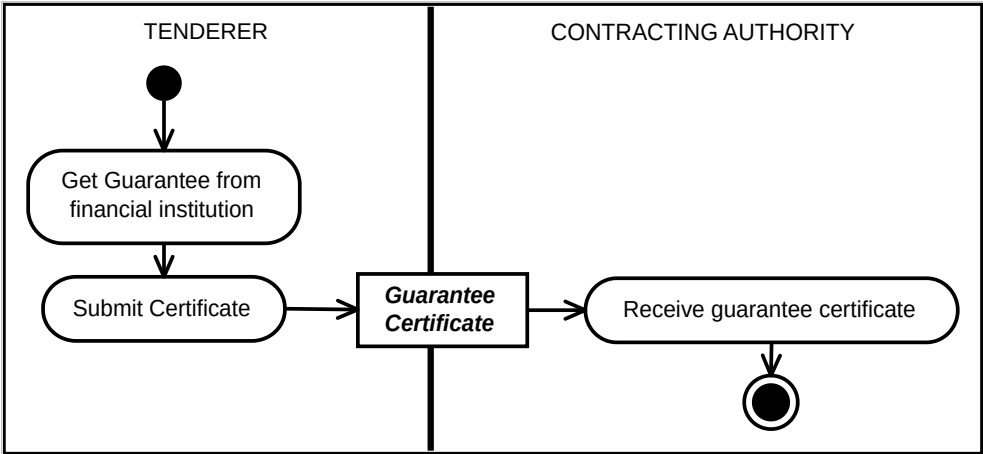
The house style proposed

The median of what the 78 artworks draw, relative to their own label size (poc-drawio/census.md).

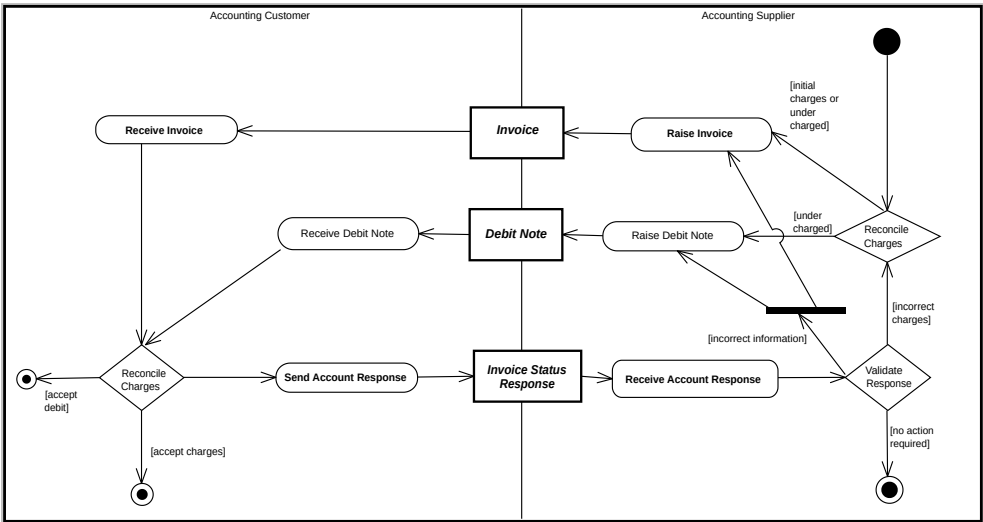
element	house style
labels	12 px Helvetica; actions bold, documents bold italic, decisions, guards and lane titles plain
lines	flows and outlines 0.10 of the label size, documents 0.16, dividers 0.11, frame 0.18
arrowheads	open, 1.35 label sizes long, the same on every flow
action	its measured size, grown only where its words would not fit; one corner radius in every action (16 px)
document	its measured size, grown only where its words would not fit; centred exactly on the lane divider it sits on
decision	its measured size, grown only around its question; 2 label sizes across when the question stands beside it
lane titles	at the same place in every figure, plain, no rule under them
start, end	draw.io's UML start and end states, 1.8 label sizes across
fork bar	0.5 label sizes thick
grey rules	none: one line per divider
rows and columns	a document level with the actions that send and receive it; boxes one above the other in a lane stacked in one column; a lane's columns evenly spaced
flows	across and down, routed round the boxes by libavoid; a decision's flows use its corners, and only a decision with more flows than corners also its slanted sides, at 45 degrees; every flow leaves or meets a diamond straight for an arrowhead's length and more

Each figure keeps its layout in its order: every element keeps its place above, below, left or right of the others, scaled so its labels come out at the house size; boxes keep their size. The boxes are then put on a grid: boxes joined by a flow across in one row, boxes one above the other in one column, a lane's columns evenly spaced. Where a box comes too close to another, or out of its lane, the figure is opened up there (section 5). The median figure grows by 5 % in width, the most by about 45 %. Flows are then routed across and down (section 4).

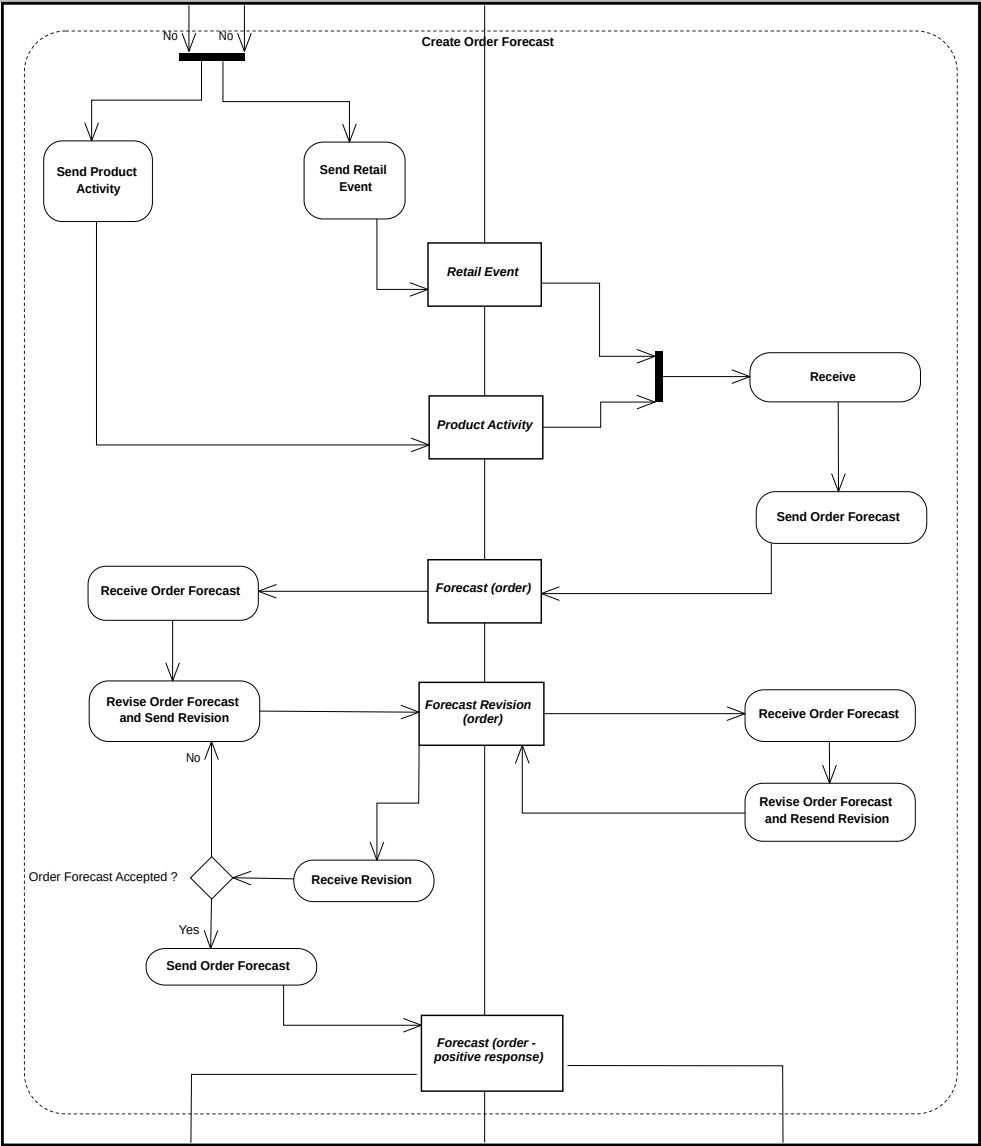
1. As published today: faithful, each at the column's width



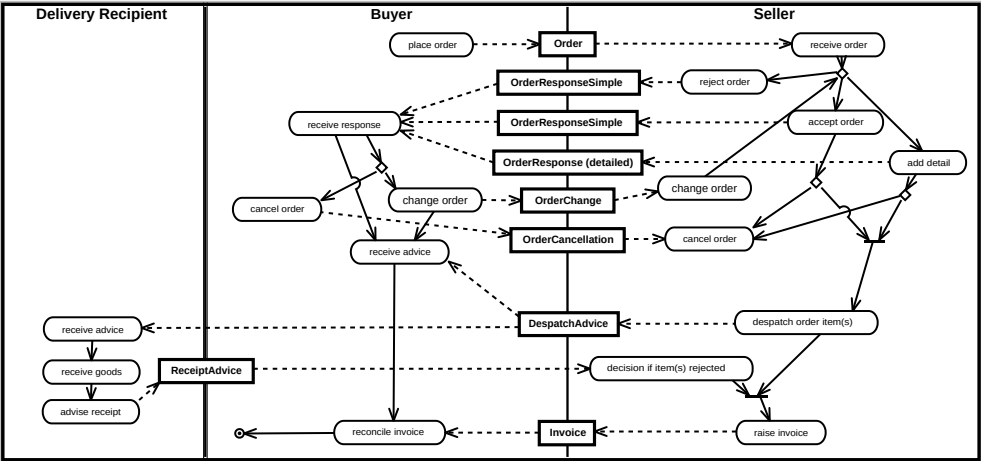
UBL-2.2-Tender-GuaranteeDeposit: 6 nodes, 5 flows. Labels print at about 9.7 pt.



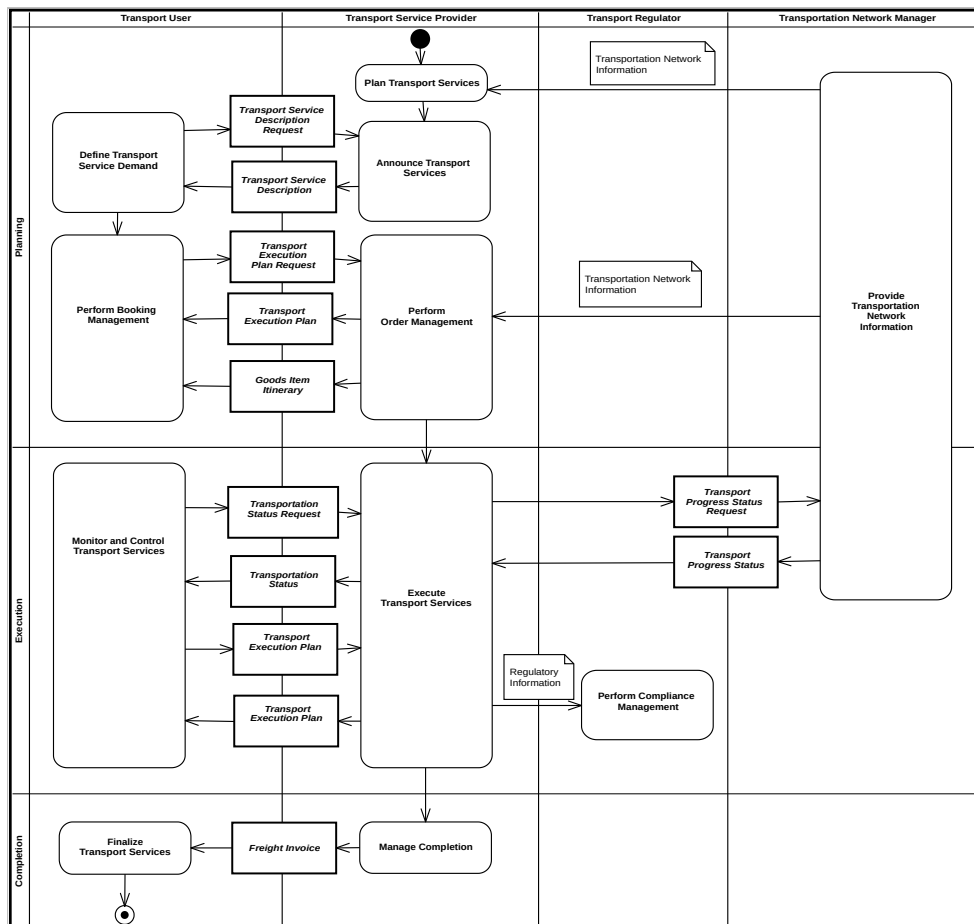
UBL-2.5-BillingwithDebitNoteProcess: 17 nodes, 20 flows. Labels print at about 4.5 pt.



UBL-2.2-CPFR-CreateOrderForecast: 18 nodes, 20 flows. Labels print at about 5.4 pt.

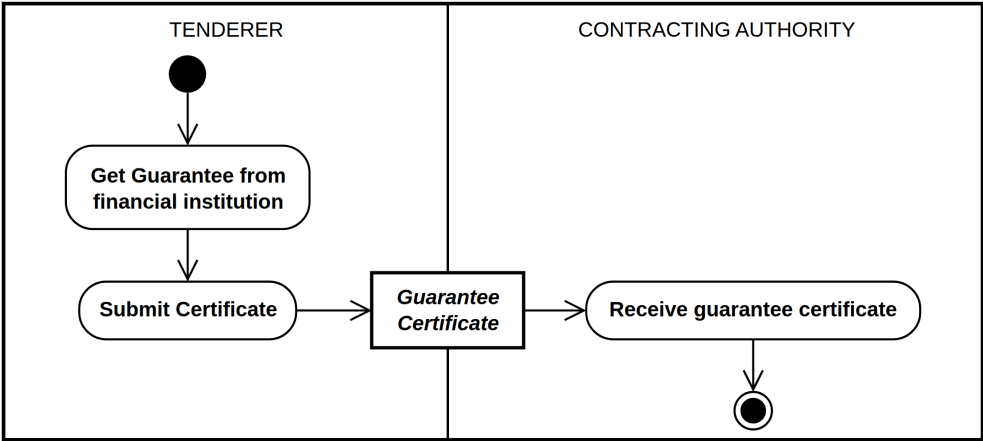


UBL-1.0-ProcurementProcess: 34 nodes, 43 flows. Labels print at about 3.9 pt.

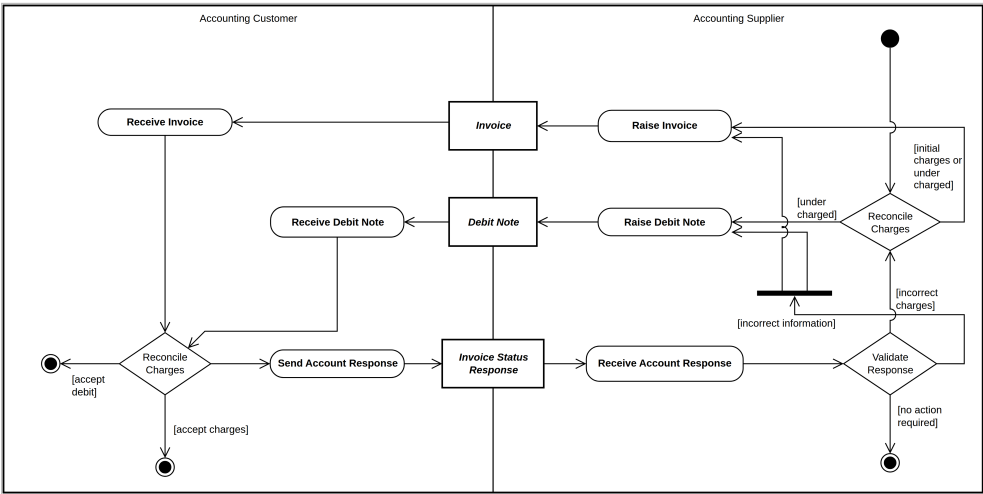


UBL-2.2-IMFM-IntermodalFreightManagementProcess: 28 nodes, 33 flows. Labels print at about 3.7 pt.

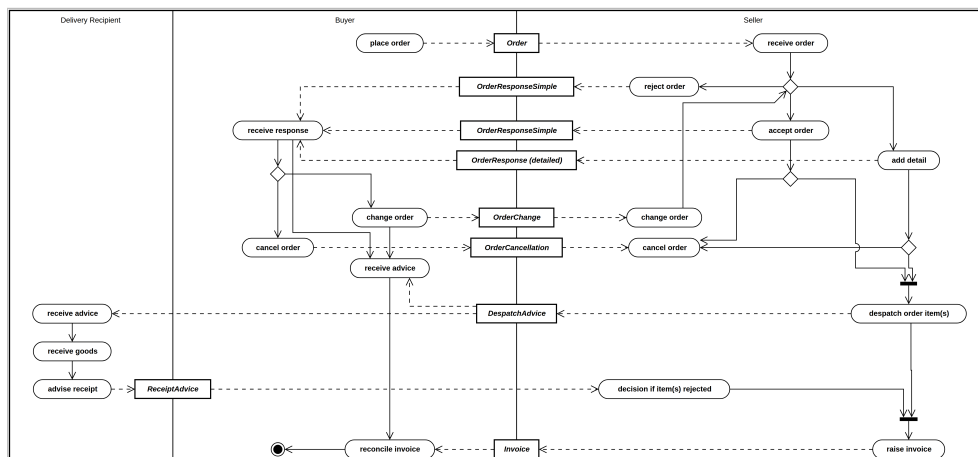
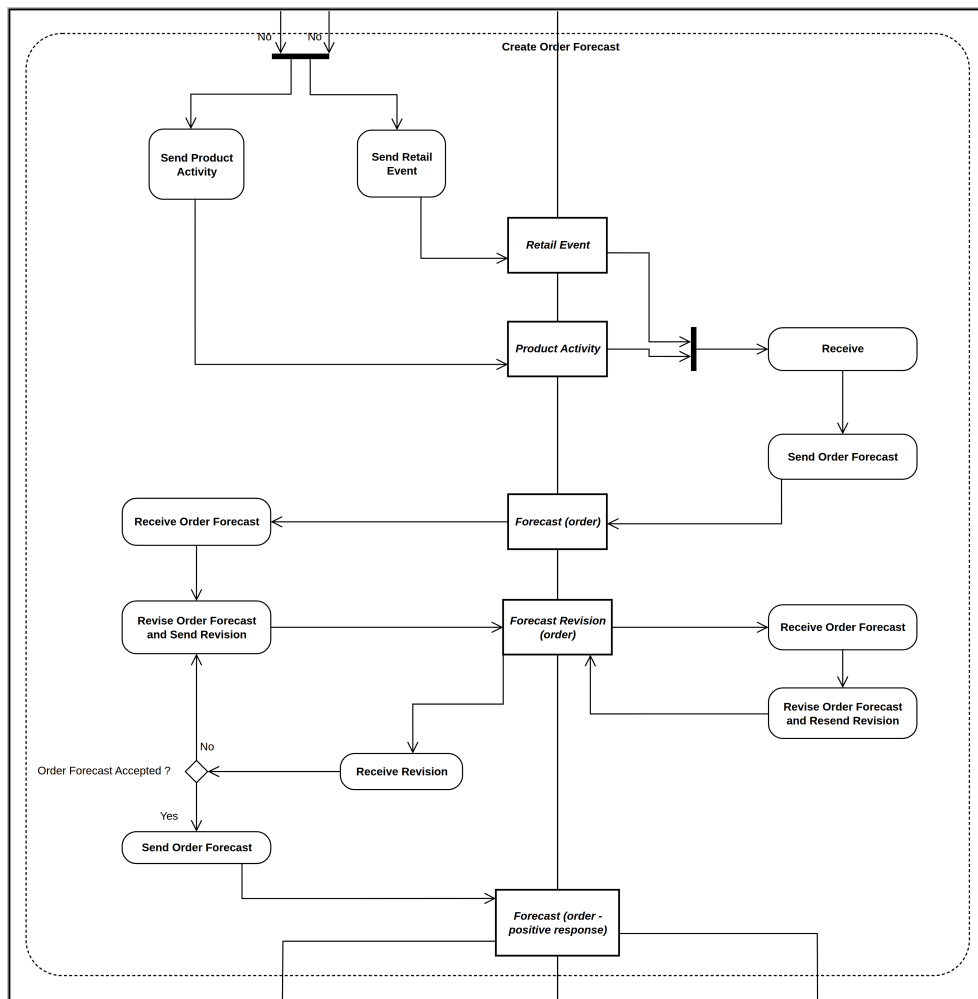
2. House style, each fitted to the column's width

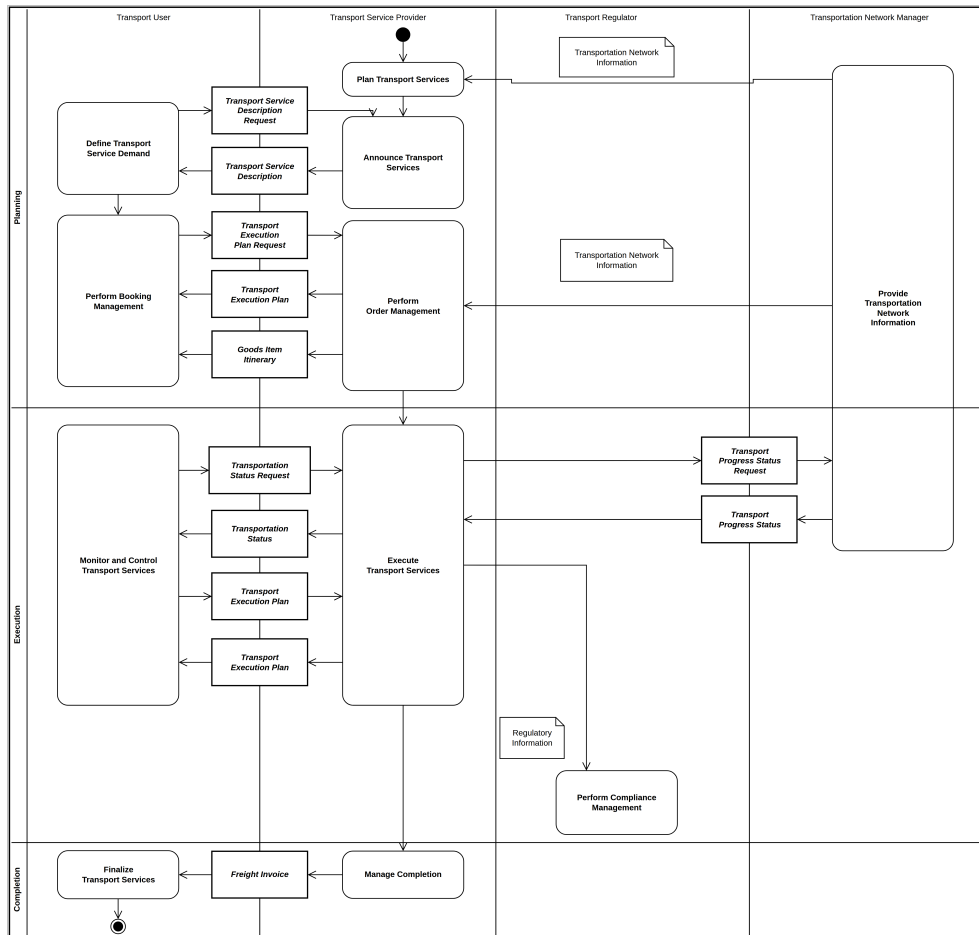


UBL-2.2-Tender-GuaranteeDeposit: labels print at 8.7 pt.



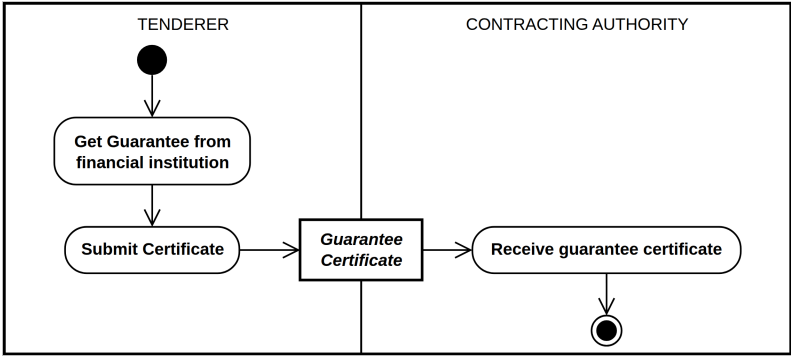
UBL-2.5-BillingwithDebitNoteProcess: labels print at 4.3 pt.



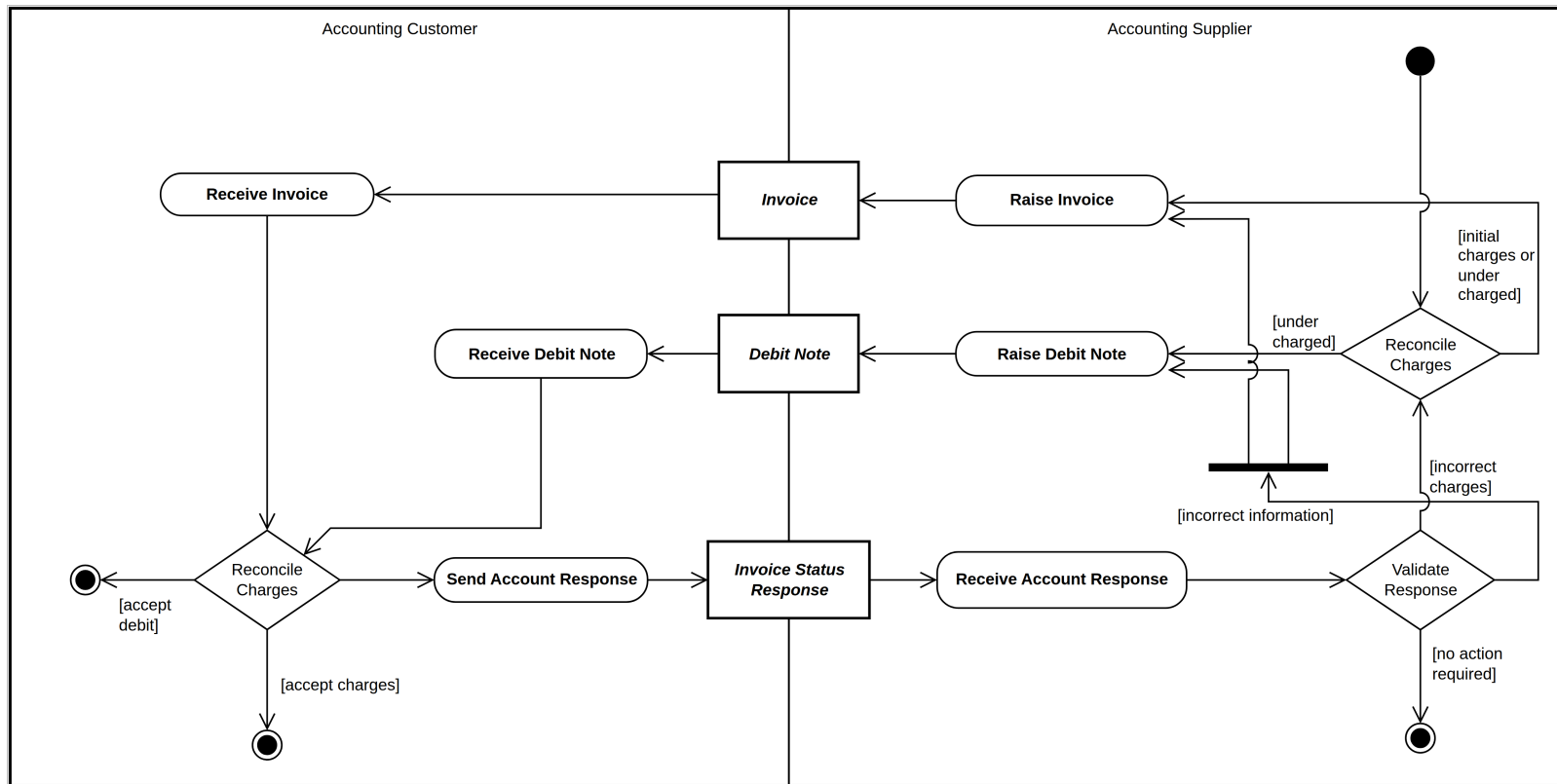


UBL-2.2-IMFM-IntermodalFreightManagementProcess: labels print at 3.4 pt.

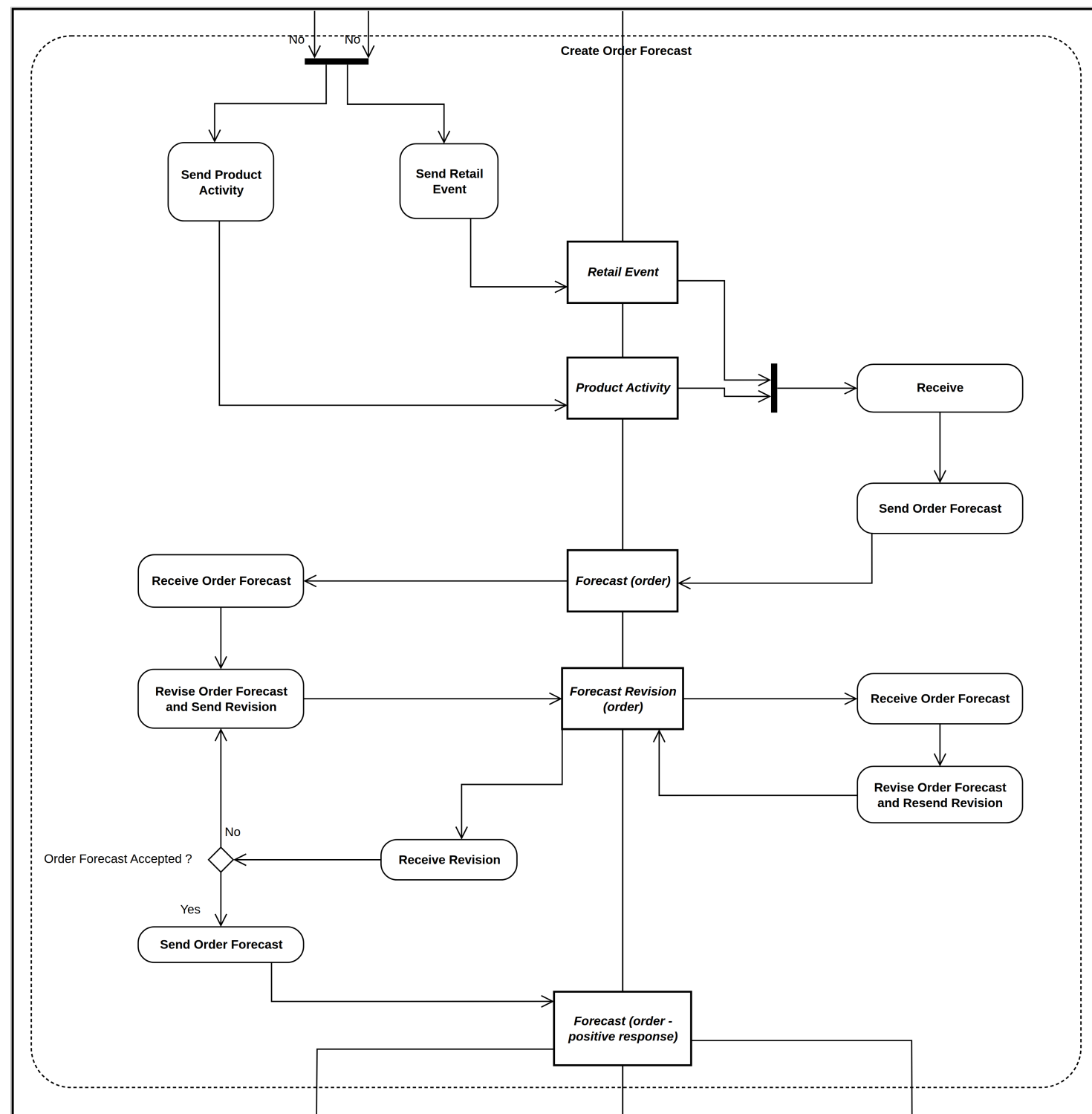
3. House style at one scale: labels 7 pt on paper in every figure



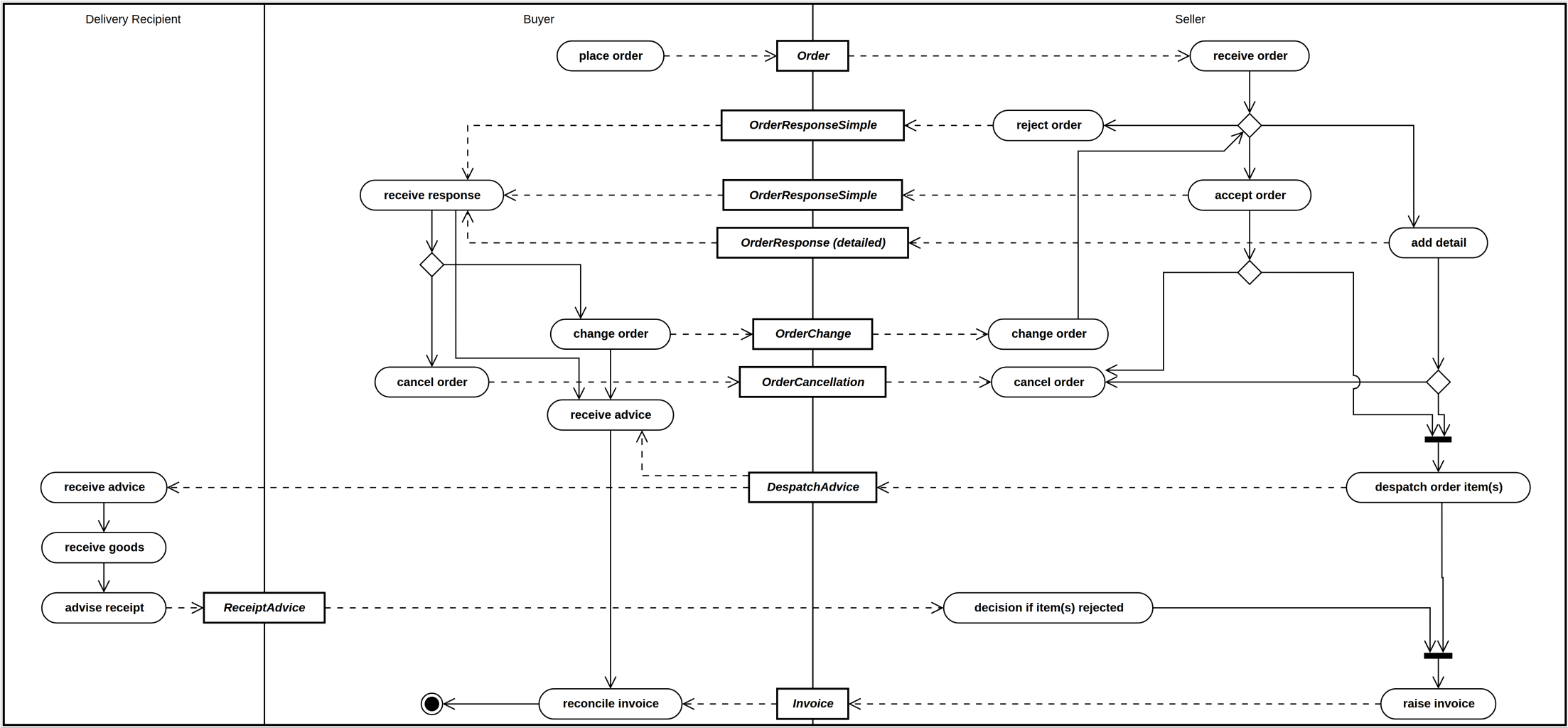
UBL-2.2-Tender-GuaranteeDeposit: 117 × 52 mm; needs A4, within the column.



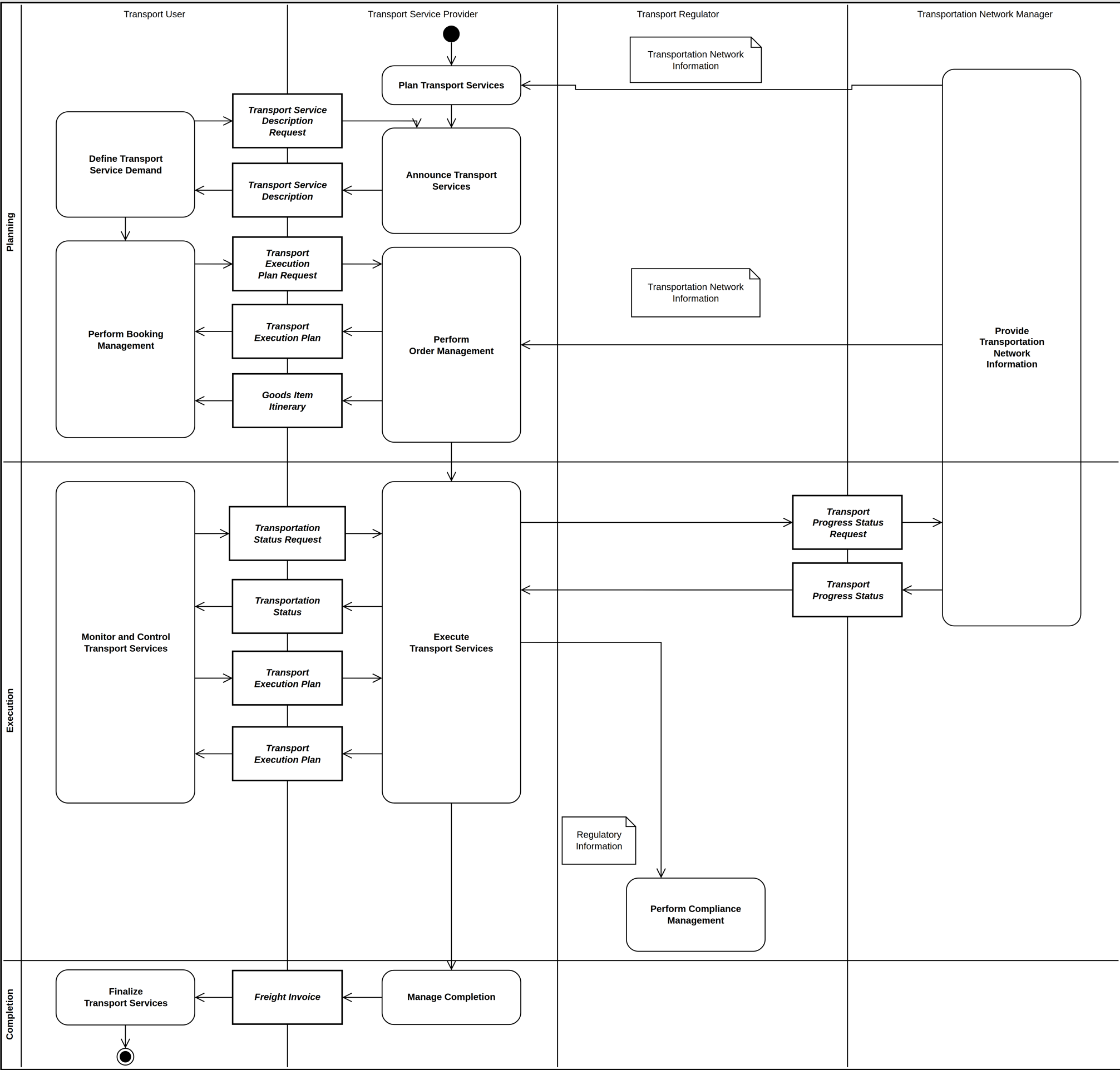
UBL-2.5-BillingwithDebitNoteProcess: 237 × 119 mm; needs an A4 landscape page.



UBL-2.2-CPFR-CreateOrderForecast: 218 × 221 mm; needs an A3 landscape page.



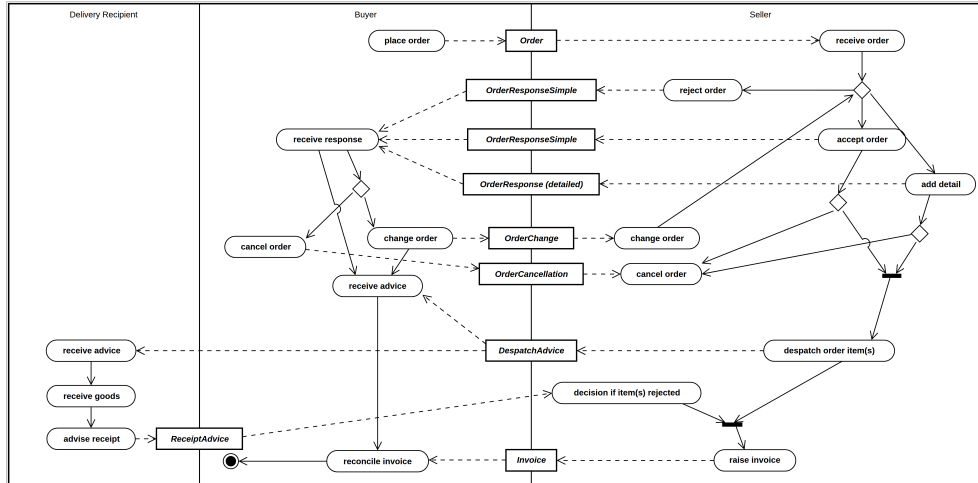
UBL-1.0-ProcurementProcess: 327 × 151 mm; needs an A3 landscape page.



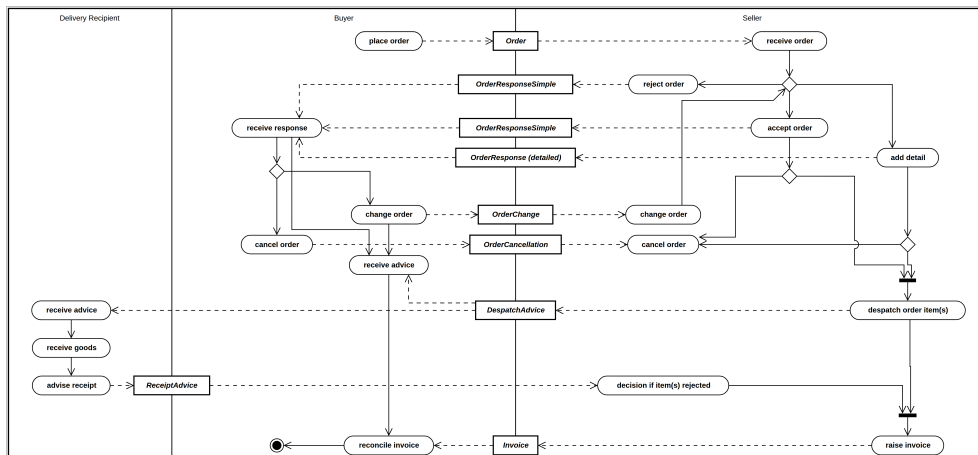
UBL-2.2-IMFM-IntermodalFreightManagementProcess: 299 × 285 mm; needs an A3 page.

4. The routing of flows: UBL-1.0-ProcurementProcess

Half the figures mix flows at an angle with flows across and down. In the house style every flow runs across and down: the boxes are first put in rows and columns with the boxes their flows join them to, and the flows are then routed round the boxes by libavoid, the router of Inkscape and Dunnart. It chooses where a flow meets a box, keeps a margin from every box, avoids bends and crossings, and spaces parallel flows apart. A decision uses its corners; only one with more flows than corners also its slanted sides, at 45 degrees. A flow in and a flow out never share a corner, and every flow leaves or meets a diamond straight for a stretch, so no arrowhead is pressed into a bend. Above, the flows at an angle kept as drawn; below, the house style.



As drawn: the layout of the artwork, flows at an angle kept.

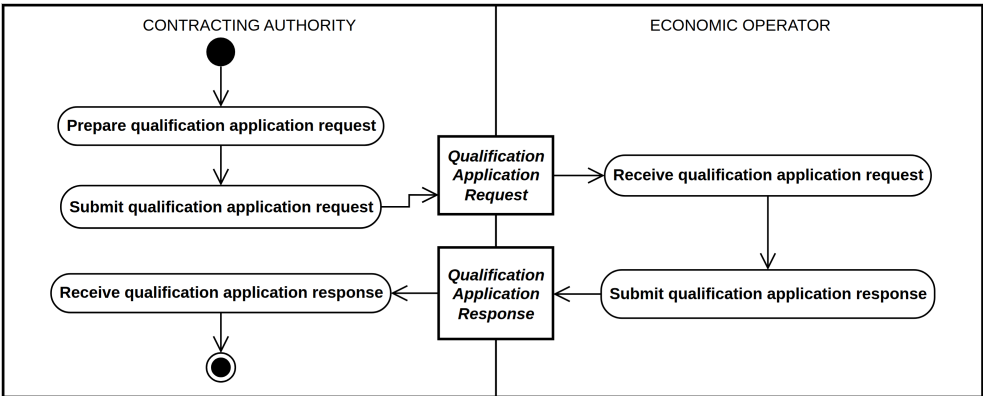


Routed across and down (house style).

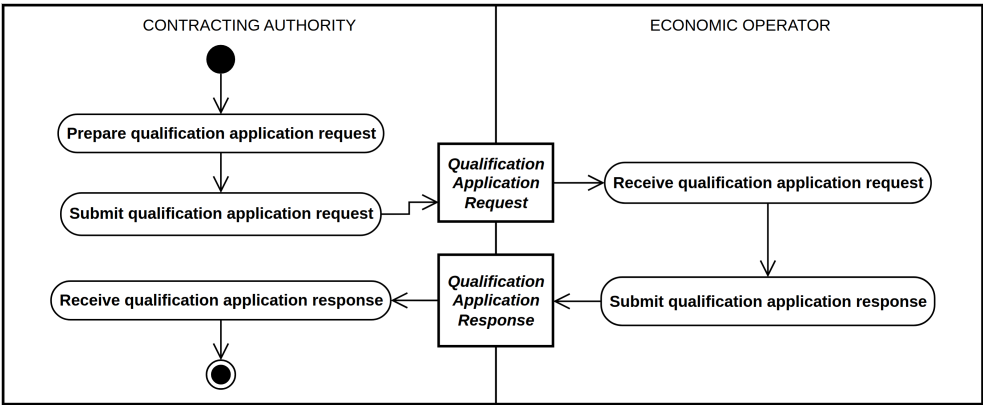
Two simpler ways were tried first and set aside: each angled flow as an L or a Z between its measured contact points, and draw.io's own router. Both ran routes through boxes and over each other. Over all 78 figures, libavoid routes 167 flows anew across and down with none through a box, and none is left unrouted.

5. Making room

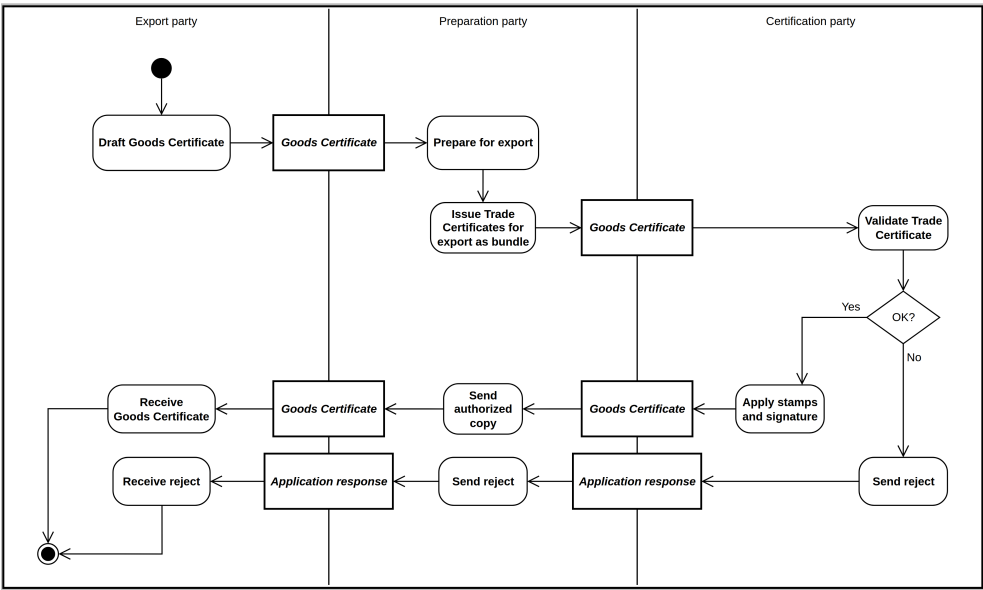
In the house style some boxes grow to fit their words, and in tightly drawn figures a flow is then too short for its arrowhead, or a box runs out of its lane. The figure is opened up where that happens: widened or lengthened at a line between the two, everything beyond moving along, so rows, columns and lanes stay aligned. Before, and after.



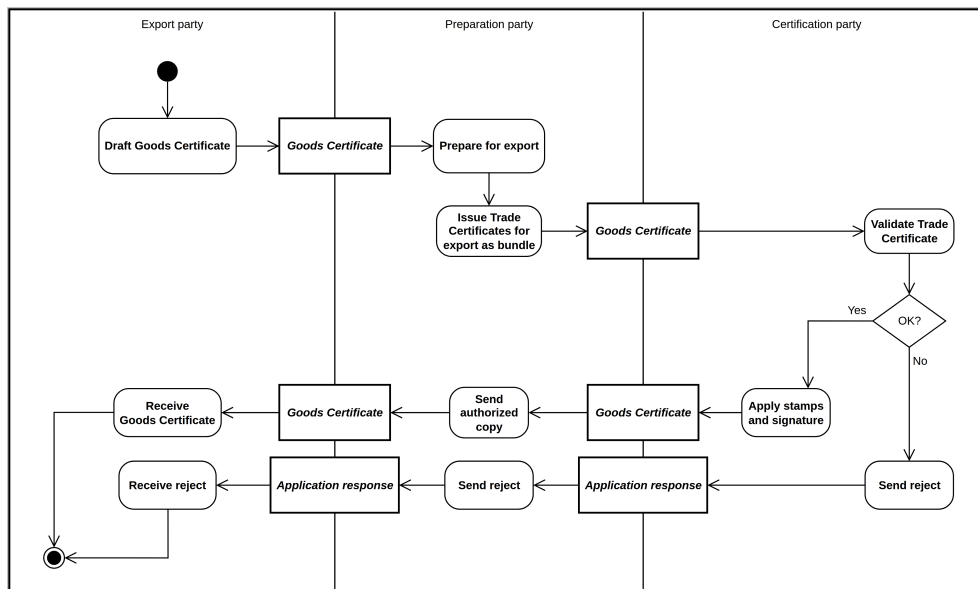
UBL-2.2-Tender-QualificationApplication: before.



UBL-2.2-Tender-QualificationApplication: after.



UBL-2.3-GoodsCertificateExportProcess: before.



UBL-2.3-GoodsCertificateExportProcess: after.